

Tutorial Quiz 8 Rubric

Q1] Parameterize the curve using
 $x=t$, $y=t^2$ for $-1 \leq t \leq 2$

$$\therefore dx = dt , \quad dy = 2t dt$$

Substituting into the integral,

$$\begin{aligned} & \int_C xy dx + (x+y) dy \\ &= \int_{-1}^2 t \cdot t^2 dt + (t+t^2)(2t) dt \\ &= \int_{-1}^2 (t^3 + 2t^2 + 2t^3) dt \\ &= \int_{-1}^2 (3t^3 + 2t^2) dt \\ &= \left[\frac{3t^4}{4} + \frac{2t^3}{3} \right]_{-1}^2 \\ &= \frac{69}{4} \end{aligned} \quad \text{--- (1)}$$

(Q2) There are multiple ways to go about this problem:

Method 1: IF F is conservative the work is same along all paths.

- Check if F is conservative using component test

$$M = x^2 + y, \quad N = y^2 + x, \quad P = ze^z$$

$$\frac{\partial P}{\partial y} = 0 = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = 0 = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = 1 = \frac{\partial M}{\partial y}$$

F is conservative.

- Find potential function f , s.t. $\nabla f = F$

$$\frac{\partial f}{\partial x} = x^2 + y \Rightarrow f = \frac{x^3}{3} + xy + g(x, y)$$

$$\frac{\partial f}{\partial y} = x + \frac{\partial g}{\partial y} = y^2 + x \Rightarrow \frac{\partial g}{\partial y} = y^2 \Rightarrow g = \frac{y^3}{3} + h(z)$$

$$\frac{\partial f}{\partial z} = \frac{dh}{dz} = ze^z \Rightarrow h(z) = ze^z - e^z + C$$

$$\therefore f(x, y, z) = \frac{x^3}{3} + xy + \frac{y^3}{3} + ze^z - e^z$$

- Evaluate $f(1, 0, 1) - f(1, 0, 0)$:

$$y(1, 0, 1) = \frac{1}{3} + 0 + 0 + e - e = \frac{1}{3}$$

$$y(1, 0, 0) = \frac{1}{3} + 0 + 0 + 0 - 1 = -\frac{2}{3}$$

$$\begin{aligned} W &= y(1, 0, 1) - y(1, 0, 0) \\ &= \frac{1}{3} - \left(-\frac{2}{3}\right) = 1 \end{aligned}$$

Since F is conservative, work is 1 along all paths — (2)

Method 2: Direct Parametric soln:

a) Parameterization: $\hat{i} + 0\hat{j} + t\hat{k}$, $0 \leq t \leq 1$

$$\therefore x=1, \quad y=0, \quad z=t$$

$$\frac{dr}{dt} = 0\hat{i} + 0\hat{j} + \hat{k}$$

Evaluating F on the curve

$$\begin{aligned} F(1, 0, t) &= (1^2 + 0)\hat{i} + (0^2 + 1)\hat{j} + t e^t \hat{k} \\ &= \hat{i} + \hat{j} + t e^t \hat{k} \end{aligned}$$

$$\begin{aligned} \text{Calculating } F \cdot \frac{dr}{dt} &= (1)(0) + (1)(0) + (t e^t)(1) \\ &= t e^t \end{aligned}$$

integrating from $t=0$ to $t=1$

$$W = \int_0^1 t e^t dt$$

using by parts integration

$$u = t, \quad du = dt$$

$$v = e^t, \quad dv = e^t dt$$

$$W = \int_0^1 t e^t dt = t e^t - e^t + C$$

$$= [t e^t - e^t]_0^1 = (e - e) - (0 - 1) = 1 \quad \text{--- (1)}$$

b) Two segment path

i) Path 1 : x -axis from $(1, 0, 0)$ to $(0, 0, 0)$

$$\text{Parameterization: } \mathbf{r}_1(t) = (1-t)\hat{i} + 0\hat{j} + 0\hat{k} \quad 0 \leq t \leq 1$$

$$\therefore \frac{d\mathbf{r}_1}{dt} = -\hat{i}$$

$$F(1-t, 0, 0) =$$

$$\begin{aligned} & ((1-t)^2 + 0)\hat{i} + (0 + (1-t))\hat{j} + 0\hat{k} \\ \Rightarrow & (1-t)^2\hat{i} + (1-t)\hat{j} \end{aligned}$$

$$F \cdot \frac{d\mathbf{r}_1}{dt} = (1-t)(-1) = -(1-t)^2$$

$$\therefore W_1 = \int_0^1 -(1-t)^2 dt = - \int_0^1 (1 - 2t + t^2) dt$$

$$= - \left[t - t^2 + \frac{t^3}{3} \right]_0^1 = -\frac{1}{3}$$

ii) Path 2: $z = x^2$, $y = 0$ from $(0, 0, 0)$ to $(1, 0, 1)$

Parameterization:

$$\mathbf{r}_2(t) = t\hat{i} + 0\hat{j} + t^2\hat{k}, \quad 0 \leq t \leq 1$$

$$\therefore x = t, \quad y = 0, \quad z = t^2$$

$$\frac{d\mathbf{r}_2}{dt} = \hat{i} + 2t\hat{k}$$

$$\begin{aligned} F(t, 0, t^2) &= (t^2 + 0)\hat{i} + (0 + t)\hat{j} + t^2 e^{t^2}\hat{k} \\ &= t^2\hat{i} + t\hat{j} + t^2 e^{t^2}\hat{k} \end{aligned}$$

$$\begin{aligned} F \cdot \frac{d\mathbf{r}_2}{dt} &= (t^2)(1) + (t)(0) + (t^2 e^{t^2})(2t) \\ &= t^2 + 2t^3 e^{t^2} \end{aligned}$$

$$W_2 = \int_0^1 \left(\underbrace{t^2}_{\frac{1}{3}} + \underbrace{2t^3 e^{t^2}}_1 \right) dt$$

$$= \frac{1}{3} + 1 = \frac{4}{3}$$

$$W = W_1 + W_2 = -\frac{1}{3} + \frac{4}{3} = 1 \quad \text{---(2)}$$